

# Customer & MVP Blueprint

Circuit.IQ — AI-Powered Virtual Physics Lab | Prepared for Anayat Zahoor | Source: Startup Validation Report, July 2026

## Executive Summary

Circuit.IQ is an AI-personalized virtual physics lab targeting India's structurally validated but under-served gap between free-but-generic tools (OLabs, PhET) and powerful-but-expensive foreign platforms (Labster). The founder has shipped a working MVP (100+ concurrent simulations, sub-second latency) but has zero direct customer validation — no interviews, no pilots, no revenue. The core task for the next 30-60 days is proving institutional demand, not building more product.

| Founder-Market Fit | TAM (India) | SAM (3-yr) | Recommendation |
|--------------------|-------------|------------|----------------|
| 6.1 / 10           | \$150M      | \$22M      | CONDITIONAL GO |

## Ideal Customer Profile (ICP)

| Attribute           | Profile                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------|
| Institution type    | Private CBSE/ICSE schools (Classes 9-12); Tier 2/3 engineering colleges & polytechnics       |
| Geography (Phase 1) | Jammu & Kashmir, Delhi NCR, + 1-2 metro clusters (Bangalore/Pune/Hyderabad)                  |
| Size                | 300-2,000 students; feels lab-infra cost pressure but has some digital budget                |
| Budget authority    | Principal/Director (schools) or HOD/Dean Academics (colleges)                                |
| Trigger conditions  | NEP 2020 compliance push, equipment shortage, low practical scores, smart-classroom branding |
| Buying behavior     | Prefers pilot-then-scale; wants free trial + visible outcome proof before commitment         |
| Disqualifiers       | Zero digital budget, no computer lab, fully offline rural schools (Phase 1 exclusion)        |

## Buyer Persona (Primary)

**Ritu Sharma — School Principal, Private CBSE School (Tier-2 city)**

"I want my students to score well in practicals, but I can't justify a new lab budget every year."

| Dimension         | Detail                                                                                       |
|-------------------|----------------------------------------------------------------------------------------------|
| Goals             | Improve practical/board scores; NEP 2020 compliance; smart-lab credentials for admissions    |
| Frustrations      | Once-a-year budget cycle; skeptical of unproven EdTech; low tolerance for complex onboarding |
| Decision criteria | Free/low-cost pilot term, clear before/after outcome data, minimal teacher training          |

Secondary personas: Dr. Anil Kumar (Engineering HOD — needs syllabus mapping + accreditation analytics) and Aditya (Class 11 student — B2B2C user via school license, not direct payer).

## Top Customer Pain Points

| # | Pain Point                                                       | Severity |
|---|------------------------------------------------------------------|----------|
| 1 | Physical lab equipment scarce/shared/damaged                     | High     |
| 2 | Existing online content is theory-heavy, low interactivity       | High     |
| 3 | Budget cycles & procurement bureaucracy breed risk-aversion      | High     |
| 4 | Free tools (OLabs/PhET) lack adaptivity, gamification, modern UX | Medium   |

| #  | Pain Point                                                          | Severity |
|----|---------------------------------------------------------------------|----------|
| 5  | No way for teachers to track per-student concept gaps in practicals | Medium   |
| 6  | Rural/low-bandwidth environments make heavy 3D tools impractical    | Medium   |
| 7  | No institutional proof points / brand trust for a new vendor        | High     |
| 8  | Long sales cycles (2-6 months) vs solo-founder bandwidth            | High     |
| 9  | Content must map to both school (CBSE/ICSE) and engineering syllabi | Medium   |
| 10 | Data-privacy concerns for minors on a new digital platform          | Medium   |

## Customer Journey: Awareness → Consideration → Purchase → Retention

| Stage         | Key Activities                                                                                |
|---------------|-----------------------------------------------------------------------------------------------|
| Awareness     | LinkedIn content (ABTalks), referrals, direct outreach to principals/HODs                     |
| Consideration | Frictionless live demo, no install, works on low-end devices; 4-8 week pilot with 1-2 classes |
| Purchase      | Simple low-friction PO/subscription suited to Indian institutional procurement                |
| Retention     | Teacher support/training, usage nudges, referral & case-study requests post-pilot             |

## Key Customer Objections

| Objection                                   | Response Strategy                                                            |
|---------------------------------------------|------------------------------------------------------------------------------|
| "We already use OLab/PhET for free"         | Position as complement: AI feedback, gamification, analytics free tools lack |
| "No budget approved this cycle"             | Offer no-cost pilot term tied to a measurable outcome                        |
| "How do we know it improves outcomes?"      | Publish a small pilot case study with before/after data                      |
| "Limited internet/device infrastructure"    | Lightweight, low-bandwidth mode emphasized upfront                           |
| "You're a student founder, no track record" | Lean into founder-problem proximity; let pilot results build credibility     |

## Key Buying Triggers

- NEP 2020 compliance pressure for demonstrable digital/practical learning
- Poor practical exam results or accreditation feedback citing weak lab exposure
- Lab equipment damage/shortage creating scheduling crunch
- Competitive pressure from a peer institution adopting a smart-lab tool
- New academic year / admission cycle budget windows

## MVP Recommendation

| Question        | Answer                                                                                                                                                                                |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Build first     | Frictionless no-install demo/trial link; low-bandwidth mode; 1-2 flagship simulations mapped tightly to Class 11-12 + first-year engineering syllabus; simple teacher usage dashboard |
| Do NOT build    | Full STEM suite (chem/bio), 3D/immersive rendering, complex LMS integrations, enterprise contract tooling, mobile app — all premature before pilot proof                              |
| Success metrics | 8-12 completed institutional interviews; 1 signed pilot; measurable engagement or pre/post concept-quiz improvement; 1+ verbal/written willingness-to-pay signal                      |

## MoSCoW Prioritization

| Priority         | Items                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Must Have        | No-install demo link, low-bandwidth mode, Class 11-12 + Yr-1 engineering physics sims, basic teacher dashboard, pilot usage tracking |
| Should Have      | Pre/post concept quiz, simple invoicing for institutions, WhatsApp/Excel-friendly reports                                            |
| Could Have       | Gamification/badges, AI misconception-detection depth, multi-language UI                                                             |
| Won't Have (now) | Chemistry/biology modules, full 3D immersive labs, LMS/API integrations, mobile native app                                           |

## Pricing Hypothesis

Position between free (OLabs/PhET) and premium-foreign (Labster, \$79-109/student/yr). Bottom-up SAM sizing implies ~\$25,000 (~\$300) average annual institutional license as a workable anchor. Recommended structure: free 4-8 week pilot → per-institution annual subscription (not per-student, to reduce procurement friction) → optional add-on tiers for analytics/accreditation reporting for colleges.

## Top 5 Risks

| Risk                                                             | Likelihood | Impact   |
|------------------------------------------------------------------|------------|----------|
| Long institutional sales cycles stall momentum                   | High       | High     |
| Free government alternative (OLabs) undercuts willingness to pay | High       | High     |
| No dedicated sales/BD resource (solo, side project)              | Med-High   | Med-High |
| Content breadth gap vs. Labster/PhET at scale                    | Medium     | Medium   |
| Data privacy/child-data compliance (DPDP Act)                    | Low-Med    | Med-High |

## 30-Day MVP Plan

| Week   | Focus & Actions                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------|
| Week 1 | Draft interview script; build target list of 30-40 schools/colleges (start J&K;); outreach via LinkedIn/email/network |
| Week 2 | Complete 8-12 structured interviews with principals/HODs/teachers; log objections & willingness to pay                |
| Week 3 | Approach top 2-3 institutions with a free, low-friction 4-8 week pilot offer; lock scope & success metrics            |
| Week 4 | Onboard teachers/students; instrument usage tracking + pre/post quiz; schedule mid-pilot check-in                     |

**Guiding principle:** No new product code this month unless it directly unblocks the pilot. The binding constraint is customer proof, not features.

## Founder Action Sheet — Top 10 Next Actions

- 1 Write a 10-question interview script for principals/HODs/teachers
- 2 Build a target list of 30-40 institutions, starting in Srinagar/J&K;
- 3 Reach out via LinkedIn (leverage ABTalks audience), email, and personal network
- 4 Conduct 8-12 structured customer interviews within 14 days
- 5 Identify the 2-3 most promising institutions from interview signal
- 6 Design a free, 4-8 week, 1-2 class pilot offer with defined success metrics
- 7 Secure at least one signed/verbal pilot commitment
- 8 Build a genuinely low-bandwidth demo mode and test on real school devices
- 9 Instrument pilot usage tracking + a simple pre/post concept quiz
- 10 Publish pilot outcome data as a case study to unlock the next institution

## Scores (0-100)

| Dimension        | Score | Rationale                                                                  |
|------------------|-------|----------------------------------------------------------------------------|
| Customer Clarity | 72    | Strong ICP/persona definition; zero direct interviews conducted yet        |
| Problem Severity | 80    | Well-corroborated by government (OLabs) and commercial (Labster) precedent |
| PMF Potential    | 58    | Clear white space, but unproven willingness-to-pay vs free alternatives    |
| MVP Readiness    | 75    | Technically strong MVP; needs pilot-instrumentation, not more features     |

## Final Verdict

■ **PROMISING BUT UNVALIDATED**

Real technical execution and a structurally validated problem, but no direct customer proof yet. Upgrade to a Strong Demand Signal requires 8-12 interviews, one committed pilot, and one written willingness-to-pay signal within 60 days. Absence of any pilot commitment after sustained 60-day outreach, or a pilot showing no measurable benefit over free tools, would indicate a pivot (coaching-institute B2B2C, white-label licensing, or freemium student-direct).